Temporal Robustness of Machine Learning-Based Flight Delay Prediction

A Case Study at Istanbul Airport

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Supervisor: Assoc. Prof. Tevfik Aytekin · Master's Thesis Defense · Istanbul, June 2026

Why Flight Delays Matter

The operational and economic scale of the problem

€13B+

Annual Economic Costs

Direct costs across European airspace: fuel overconsumption, crew overtime, passenger compensation (EUROCONTROL CODA, 2024)

620K+

Annual Flights at IST

Istanbul Airport handled over 620,000 flights in 2023, ranking #1 in Europe by aircraft movements (EUROCONTROL, 2024)

51%

Hub Concentration

Turkish Airlines share of IST departures — dense rotation schedules amplify delay propagation across linked flights (DHMI, 2024)

76M+

Passengers Affected

IST served over 76.5 million passengers in 2023. Missed connections and extended travel times compound the economic toll

Why Istanbul Airport?

An operationally complex environment for temporal robustness research

#1

Busiest in Europe

by aircraft movements, 2023 (EUROCONTROL)

620K+

Annual flights

handled at IST/LTFM

120+

Countries connected

via international routing

51%

TK departure share

of all IST departures

Why This Airport?

TK's dense rotation schedule means a single disruption can cascade across an entire departure bank before recovery protocols activate.

51% hub concentration creates tightly coupled flight sequences, a consequential environment for studying delay propagation dynamics.

120+ country network introduces heterogeneous carrier behaviour, seasonal patterns, and operational conditions.

Results may offer methodological insights applicable to similar large hub airports, within the constraints of this single-airport study.

Note: findings are specific to this airport and study period and should be interpreted within these constraints.

The Core Problem

Why offline performance metrics may substantially overestimate real-world effectiveness

Traditional Evaluation

- Random train-test splits — temporal periods artificially mixed
- Training and test data drawn from the same calendar months
- Assumes feature–target relationships are stationary over time
- Often reports high AUC but may reflect interpolation, not forecasting

VS

Real-World Deployment

- Future data differs from historical data; distributions shift
- Seasonal operational changes alter feature–delay relationships
- Concept drift: the statistical relationship between inputs and target evolves over time
- Performance may degrade substantially after deployment

Key problem: High offline performance does not guarantee reliable deployment. The temporal robustness of flight delay prediction models remains insufficiently examined in the literature.

Research Questions

Four questions guiding the investigation

RQ1 Performance

How accurately can ML models predict flight delays at IST, and what is the relative influence of different feature categories?

RQ2 Generalization

Do predictive relationships learned from an autumn period remain valid when models are evaluated on subsequent winter operations?

RQ3 Mechanism

What mechanisms drive temporal performance degradation, and which modelling strategies can mitigate it?

RQ4 Traffic

Does the inclusion of surface road traffic variables provide meaningful predictive signal for departure delays at IST?

Literature Review & Research Gap

What exists, what is missing, and how this thesis contributes

What Exists

- Statistical and ML models widely applied to delay prediction
- Random Forest, XGBoost, deep learning variants tested
- Strong reported performance under random-split evaluation
- Focus primarily on maximising predictive accuracy

What Is Missing

- Limited temporal validation across chronologically separated periods
- Seasonal concept drift rarely examined explicitly
- Delay propagation dynamics across seasons underexplored
- Real deployment reliability of static models uncertain

What This Thesis Does

- Implements strict temporal validation (train: Oct–Nov, test: Jan–Feb)
- Examines seasonal variation in predictive relationships
- Quantifies delay propagation dynamics across seasonal regimes
- Evaluates contribution of surface traffic data (empirical test)

Key Gap: Relatively few published studies evaluate models using chronologically separated multi-month training and testing periods. As a result, model robustness under temporal distribution shifts and the mechanisms that drive degradation remains largely unexplored in the single-airport prediction literature.

Concept Drift: A Central Theoretical Challenge

Why a model trained in one period may fail in another, even when overall delay rates stay stable

Definition: Concept drift refers to the change over time in the statistical relationship between input features and the target variable. When the data-generating process evolves, a trained model's learned associations may no longer hold, causing predictive performance to degrade without any obvious change in the raw input values.

AUTUMN (Train)

High delay rate in prev hour
→ predicts: delayed
→ correct ✓

Relationships are learned and stable

WINTER (Deploy)

Lower prev-hour delay rate
→ model still predicts: delayed
→ incorrect ✗

Input values shift; model uses old rules

In this study: the feature distributions shift across seasons while overall delay prevalence remains relatively stable (79.5% → 77.6%), suggesting concept drift in the feature–target relationship rather than a simple base-rate change.

Delay Propagation: The Core Mechanism

Why airports are interconnected systems, not collections of independent flights



Shared Aircraft

Late-arriving aircraft delays its next rotation, no spare capacity to absorb the disruption.

Shared Crews

Crew schedules are tied between flights. Duty-time limits may require costly reserve activation.

Shared Gates

Delayed departures occupy gates past scheduled block time, blocking subsequent boarding.

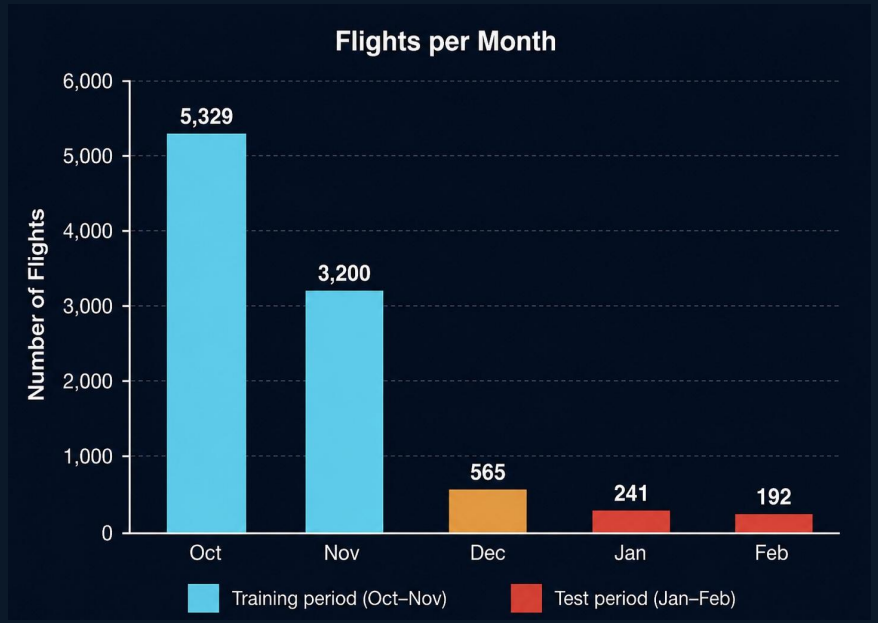
Hub Cascades

Dense departure banks at major hubs amplify propagation across multiple interconnected flights.

Dataset Overview

Multi-source operational data: October 2025 – February 2026

Airport	Istanbul Airport (IST / LTFM)
Study Period	October 2025 – February 2026 (5 months)
Final Dataset	9,527 flights (from 42,623 raw API records)
Prediction Target	Departure Delay ≥ 15 min (binary)
Overall Delay Rate	79.4% delayed | 20.6% on-time
Data Sources	Flight records · Weather (hourly) · Surface traffic
Train / Test Split	Oct–Nov (8,529) → Jan–Feb (433)



79.4% delay rate is high relative to typical airport statistics. It reflects: (1) the 15-minute threshold capturing minor deviations, (2) Istanbul's operational conditions, and (3) possible selection bias from excluding flights lacking actual departure timestamps.

Data Quality and Missingness

A critical transparency step: from 42,623 raw records to 9,527 analysable flights

42,623
Raw API records retrieved

→ *AviationStack API*

9,814
Records with actual departure timestamps

→ *−77% (future/cancelled flights, API gaps)*

9,609
After delay outlier removal (>720 min)

→ *−82 extreme outliers (data errors / diversions)*

9,527
Final analytical dataset

Weather imputation + bias correction

MCAR Assumption (Not Formally Tested)	77% attrition from raw records primarily reflects future scheduled flights and API gaps. Exploratory checks found no obvious systematic patterns, but a formal MCAR test was not conducted. The possibility of selection bias cannot be excluded.
Surface Traffic Data: Averaged Profile	Due to data access constraints, traffic features use 24-hour averaged profiles (not actual daily values). This means day-to-day traffic variability is absent — the test of RQ4 therefore captures only circadian patterns, not real-time road conditions.
Weather Imputation (4-hour window)	Missing weather data (78% of hours) was imputed via temporal nearest-neighbour matching within a conservative 4-hour window. Autocorrelation at 4h lag is $\rho > 0.73$ across all variables. Relative imputation error < 8% for continuous variables.

Feature Engineering

Five feature categories constructed from operational, meteorological, and traffic data

Historical Delays

- prev_hour_delay_rate — strictly chronological (no leakage)
- avg_delay_last_24h — rolling 24-hour delay rate
- airline_7d_delay_rate — carrier-specific 7-day rolling rate

Weather

- Temperature, Humidity, Pressure, Wind Speed
- Weather category (Clear / Clouds / Rain / Snow)
- Hourly observations matched within 4-hour tolerance

Airline Identity

- One-hot encoded carrier identifier
- Captures persistent carrier-level reliability differences
- Airlines < 30 flights aggregated to 'Other'

Surface Traffic

- road_avg_speed on airport access routes
- road_vehicle_count, road_congestion_index
- ⚠️ 24h averaged profiles — no day-specific variability

Temporal

Hour of day (0–23) · Day of week (0–6) · Captures circadian departure bank patterns · Traffic features use 24h averaged profiles — day-level variability absent

Temporal Leakage Prevention

A critical validity safeguard — ensuring historical features do not encode the target

✗ INCORRECT APPROACH

```
prev_hour_delay_rate =
(Flight A + B + C + D) / 4
```

Flight D is the current flight being predicted — including it creates circular dependency. The model 'sees' whether this flight is delayed. Training accuracy inflates artificially then collapses at deployment.

Result: spuriously high training accuracy

✓ CORRECT APPROACH

```
prev_hour_delay_rate =
(Flight A + B + C) / 3
```

Only flights with scheduled departure BEFORE flight D are included. Implemented via strict chronological iteration — not vectorised computation, which is susceptible to inadvertent look-ahead.

Validated: feature mean diverges from overall delay rate

Why it matters: Temporal leakage is among the most common validity threats in aviation ML. Without strict leakage prevention, the AUC achieved could reflect circular dependency rather than genuine predictive signal. This study's protocol explicitly addresses this gap in the published literature.

Modelling Approach & Evaluation Strategy

Three models spanning the interpretability–complexity spectrum, two validation protocols

Logistic Regression

Linear baseline

Interpretable coefficients · L2 regularisation · Class-weight balanced

Random Forest

Ensemble (bagging)

Bootstrap aggregation · Random subsampling · 30-iteration random search

XGBoost

Ensemble (boosting)

Sequential error correction · L1/L2 regularisation · SHAP-compatible

Why not deep learning? 9,527 flights insufficient for DL capacity advantages. Interpretability critical for operational contexts. Ensemble methods match DL performance at this data scale (prior literature).

Random-Split Evaluation

- 75/25 stratified split (temporal periods mixed)
- Comparable to published benchmark conditions
- may overestimates deployment performance

Temporal Validation

- Train: Oct–Nov 2025 (8,529 flights)
- Intermediate: Dec 2025 (565 flights)
- Test: Jan–Feb 2026 (433 flights)
- Simulates genuine forecasting across seasons

Primary metric: ROC-AUC (threshold-independent, robust to 79.4% class imbalance)

Handling Class Imbalance

79.4% delayed vs 20.6% on-time — strategies and rationale

79.4%

Delayed class — naive accuracy would be misleading as a metric

ROC-AUC as Primary Metric

Threshold-independent. Measures ranking ability across all decision thresholds. Robust to class imbalance — a majority-class guesser scores exactly 0.5.

Applied ✓

Class-Weight Balancing

Logistic Regression uses class_weight='balanced', inversely scaling loss by class frequency. XGBoost uses scale_pos_weight = (n_negative / n_positive) ≈ 3.85.

Applied ✓

SMOTE — Considered & Rejected

Synthetic oversampling considered but rejected: SMOTE on temporal data risks generating synthetic samples that violate the chronological ordering of the leakage-prevention protocol.

Rejected — leakage risk

Why accuracy fails here: a model that predicts 'delayed' for every flight achieves 79.4% accuracy, yet has zero discriminative power. AUC correctly scores this at 0.5.

Hyperparameter Tuning

Search space, strategy, and AUC gains for Random Forest and XGBoost

Strategy: Randomized Search CV · 30 iterations · 5-fold stratified cross-validation within training period · Scoring metric: ROC-AUC

Random Forest — Key Parameters

Parameter	Search Range	Best Value
n_estimators	100 – 500	300
max_depth	5 – 30 / None	20
min_samples_split	2 – 20	5
min_samples_leaf	1 – 10	2
max_features	sqrt / log2 / 0.3	sqrt
class_weight	balanced / None	balanced

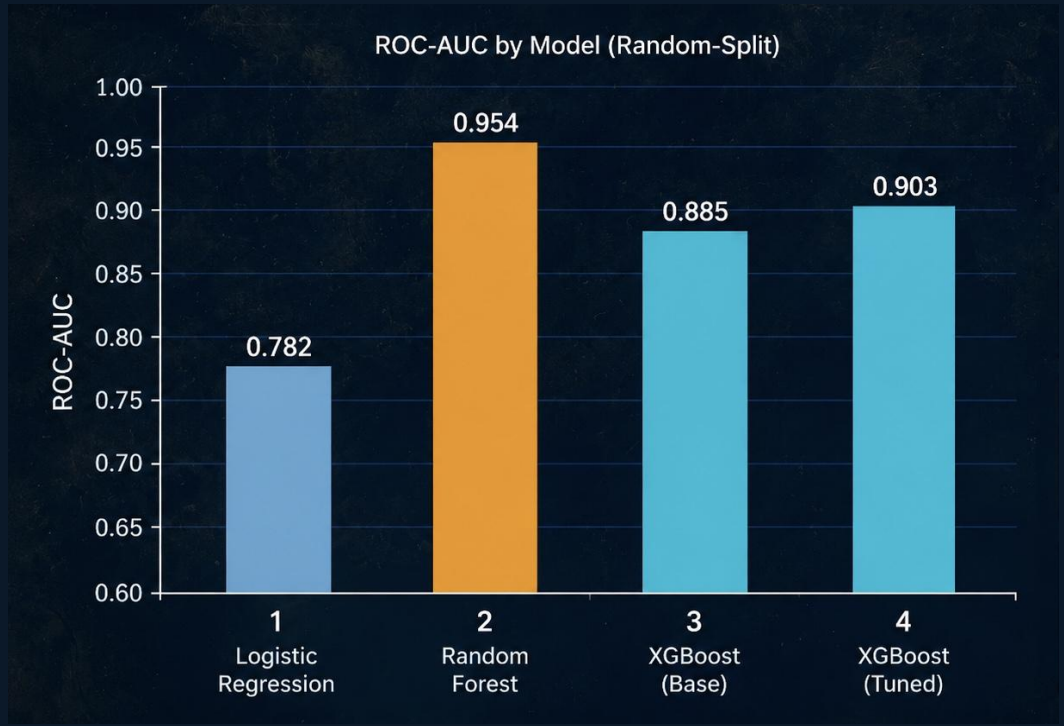
XGBoost — Key Parameters

Parameter	Search Range	Best Value
n_estimators	100 – 500	400
max_depth	3 – 10	6
learning_rate	0.01 – 0.3	0.05
subsample	0.6 – 1.0	0.8
colsample_bytree	0.6 – 1.0	0.8
scale_pos_weight	1 / 3.85	3.85

AUC gain from tuning: **XGBoost default → tuned: 0.881 → 0.903 (+2.2pp)** **RF default → tuned: 0.931 → 0.954 (+2.3pp)** *[random-split evaluation]*

Baseline Results: Random-Split Evaluation

Phase 2 dataset (9,527 flights) — results under conventional evaluation conditions



0.903 — **XGBoost Tuned — Best AUC**

Under random-split evaluation

0.941 — **Recall**

91.4% of delayed flights correctly identified

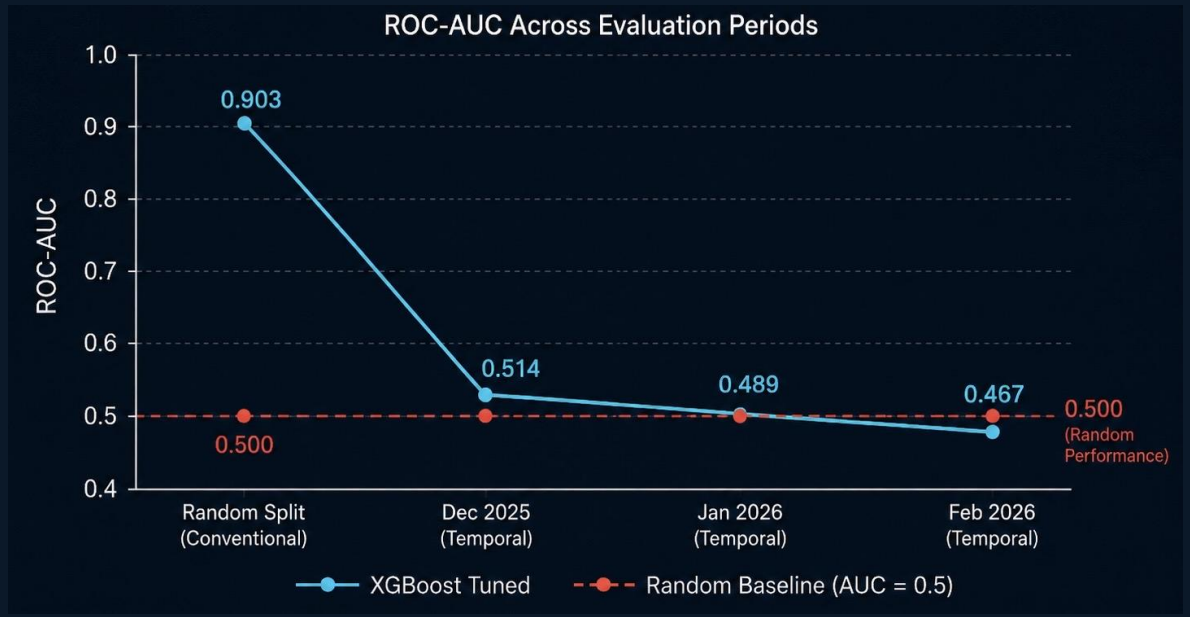
0.865 — **Precision**

Well-calibrated predictions for majority class

⚠️ These results are from random-split evaluation (temporal periods mixed) — they reflect interpolation within a known distribution, not genuine forecasting. Temporal validation is examined next.

Performance Under Temporal Validation

Training on Oct–Nov 2025 · Testing on Dec 2025, Jan–Feb 2026



0.903

Random-Split AUC

Conventional evaluation — not deployment-realistic

0.514

December (−1 mo)

Performance begins to degrade within 1 month

0.474

Jan–Feb (−3–4 mo)

Below random baseline — predictions are inverted

All three model architectures (LR, RF, XGBoost) show consistent collapse, suggesting a distributional rather than architectural explanation. Overall delay rates remained relatively stable across periods (76.8–79.5%), indicating that the failure reflects changes in feature–target relationships, not a change in base rates.

The December Paradox

Stronger propagation signal, yet AUC collapses within one month. Why?

Oct–Nov (Train)

+7.07pp

Contagion effect

Model learns: high prev-hour delay rate → predict delay. Rule fits autumn dynamics.

December (Transition)

+17.10pp

Contagion effect

Propagation strengthens but the MODEL wasn't trained on this dynamic. It saw +7pp, not +17pp. The magnitude is out of distribution.

Jan–Feb (Test)

−7.78pp

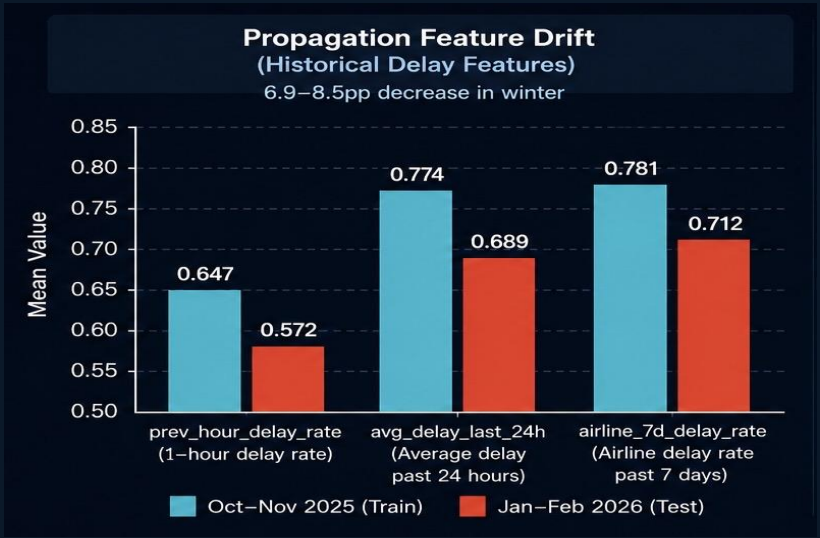
Contagion effect

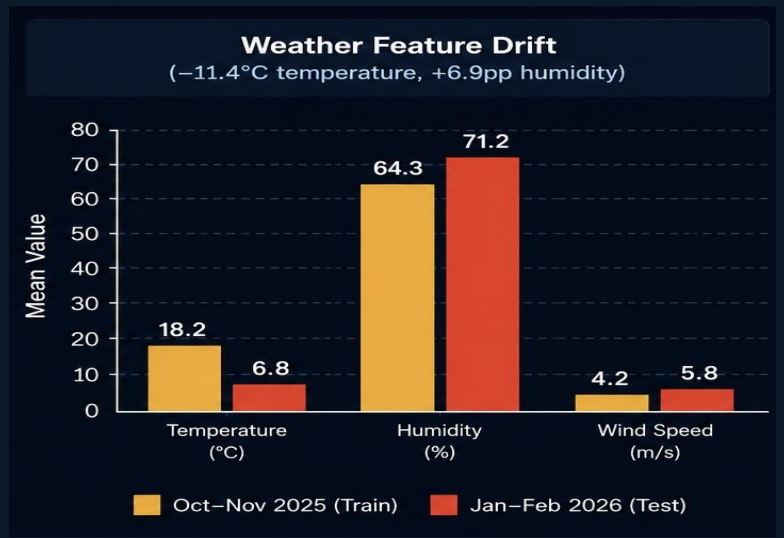
Relationship reverses. Prior delays now negatively predict subsequent delays. Model's learned rule becomes adversarial.

Key insight: December's AUC of 0.514 is not a slow drift, it's an immediate collapse. A model trained on +7pp autumn dynamics cannot generalise even when the direction of propagation stays positive. The magnitude shift alone is sufficient to destroy discriminative power.

Distribution Shifts Between Training and Test Periods

Feature-level evidence of the conditions underlying temporal performance degradation





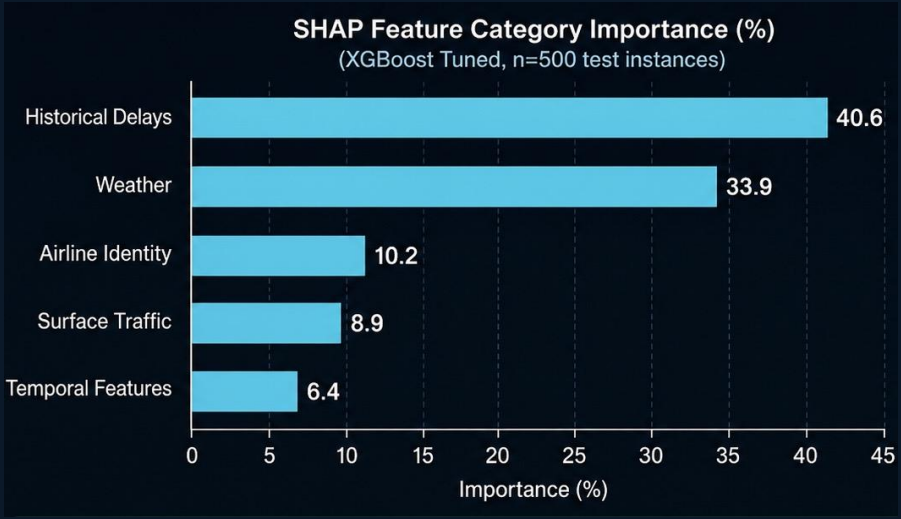
The Distributional Paradox:

Historical delay features decreased by 6.9–8.5 percentage points between training and test periods, while overall delay prevalence remained relatively stable (79.5% → 77.6%). The model was trained to rely heavily on propagation features, but those features shifted substantially in winter. These results are consistent with concept drift in the feature–delay relationship across seasonal regimes.

SHAP Feature Importance Analysis

Identifying what the model relies on — and why that creates temporal vulnerability

SHAP (SHapley Additive exPlanations): assigns each feature a contribution value for each individual prediction, grounded in cooperative game theory. Provides both global importance rankings and local per-prediction explanations.



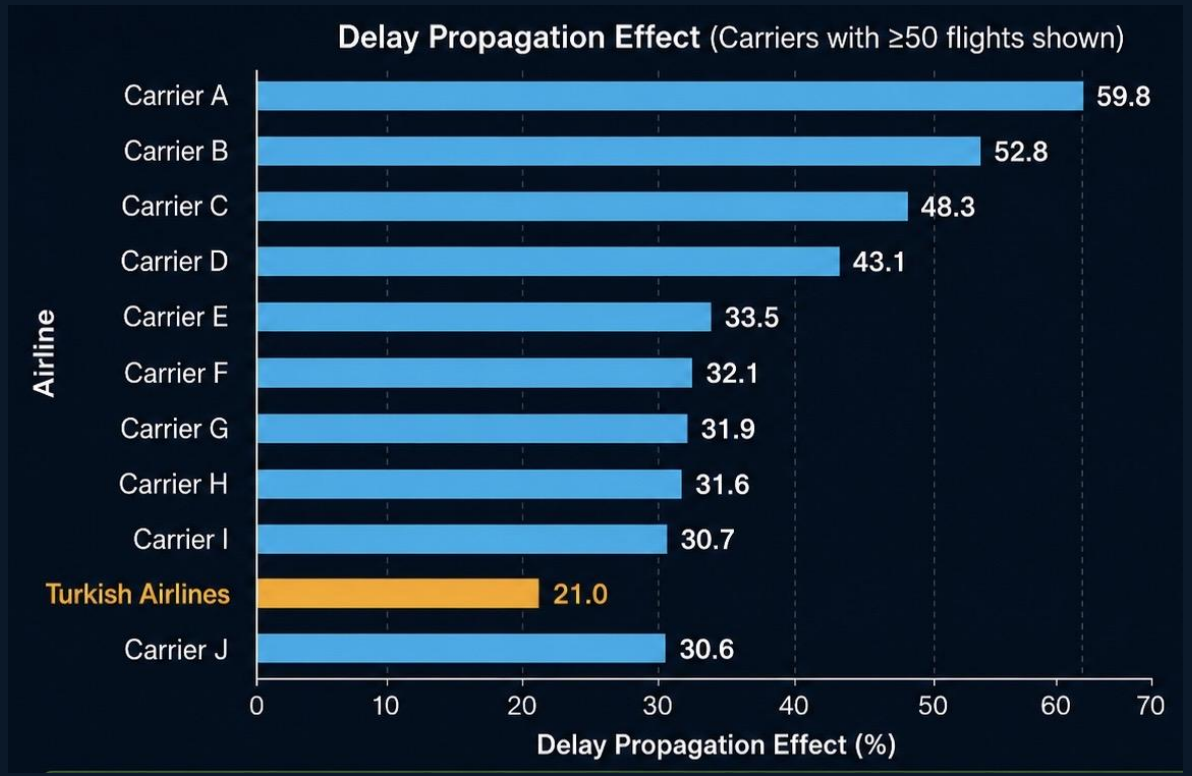
Top Individual SHAP Values

`airline_7d_delay_rate`	~0.50
`avg_delay_last_24h`	0.42
`prev_hour_delay_rate`	0.30
`wind_speed`	0.27
`pressure`	0.26

The Critical Link:
The most important predictive features are precisely the historical delay propagation variables (40.6% of total SHAP importance). These same features exhibited the largest distributional shifts between training and test periods (6.9–8.5pp decrease). The observed temporal degradation is therefore consistent with though not definitively caused by changes in the behaviour of the model's dominant predictors.

Delay Propagation: Airline-Specific Heterogeneity

Comparing delay contagion effects across carriers (≥50 flights, all periods)



+7.2pp

Overall Effect

Across all 9,527 flights, $p < 0.0001$

+59.8pp

Aegean Airlines

89.2% delayed after a delay vs 29.4% after on-time

+21.0pp

Turkish Airlines

3,498 flights, confirms systemic hub propagation

Airline names anonymised where not identifiable from public data.

Surface Traffic Analysis: Limited Predictive Value

Assessing RQ4: does road traffic add meaningful signal for departure delays?

⚠️ Critical data limitation: Traffic variables use 24-hour averaged profiles (not actual daily values). Day-to-day variability from accidents, weather, or events is absent. This study tests whether the typical circadian traffic pattern correlates with delays, a more restricted test than real-time traffic integration.

r = 0.03

Pearson Correlation

Mean absolute correlation across all three traffic features (avg speed, vehicle count, congestion index). Near-zero across all time-of-day strata.

Negligible

−0.002

Ablation Study (ΔAUC)

AUC change when all traffic features removed from full model. Removing traffic has essentially no measurable impact on predictive performance.

Negligible

8.9%

SHAP Importance

Combined contribution across all traffic features, the lowest of any feature category. Weather alone contributes 33.9%; historical delays 40.6%.

Lowest

Why traffic may not contribute:
Airlines build schedule slack into crew reporting requirements and check-in deadlines to absorb typical road variability. Airside operational constraints likely dominate.

Interpretation (RQ4):
Results suggest the averaged circadian traffic profile contributes negligible predictive signal. Whether real-time, day-specific traffic data would yield a different result remains an open question for future research.

Robustness Experiments: All 6 Strategies

What each strategy does, and why some work better than others

1. Global XGBoost (Baseline) — AUC: 0.474

Trained Oct–Nov, tested Jan–Feb with no adaptation. Establishes the degradation floor. All subsequent strategies are evaluated against this.

2. Expanding Window — AUC: 0.566

Retrains on all available data up to the test month (Oct through Dec). More data, but the Oct–Nov autumn signal still dominates. Distributional shift persists.

3. Sliding Window (2 months) — AUC: 0.573

Trains on the 2 most recent months before the test period. Reduces autumn over-representation slightly. Marginal improvement, still not enough to capture regime change.

4. Sliding Window (1 month) — AUC: 0.569

Trains on December only. Highly limited training data (565 flights). Insufficient sample for stable model fitting despite better seasonal alignment.

5. Season-Aware Random Forest — AUC: 0.599

Training restricted to autumn months only (Oct–Nov), with explicit seasonal labels. Seasonal specialisation helps, the model is no longer confused by mixing seasonal regimes.

6. Propagation Removal (Season-Aware) — AUC: 0.624

Removes all 3 historical delay propagation features within the season-aware framework. Best result: eliminating adversarial features recovers AUC to 0.624.

Caveat: Strategy 6 was evaluated within the season-aware framework (Strategy 5). The individual contributions of feature removal vs. seasonal specialisation cannot be fully disentangled.

What 'Season-Aware' Means Operationally

Defining seasons, restricting training, and the design choices involved

Definition: A season-aware model is trained exclusively on data from the same meteorological and operational season as the intended deployment period rather than on all available historical data regardless of season.

How seasons are defined

Calendar-based: Oct–Nov = Autumn, Dec = Transition, Jan–Feb = Winter

Operationally motivated: aligns with IATA seasonal scheduling periods (summer: late Mar–Oct, winter: Nov–Mar)

Validated by the observed propagation reversal: the +7pp → −7.78pp swing confirms distinct operational regimes

What training data is used

Season-Aware RF (Strategy 5): trains on Oct–Nov only (8,529 flights)

No cross-season mixing, December and winter data deliberately excluded from training

Consequence: smaller training set, but higher distributional alignment with the test period

Design choices & limitations

Season boundaries are calendar-defined, not data-driven, a cluster-based approach could be more principled

Only one seasonal transition observed, cannot verify the season definition generalises across years

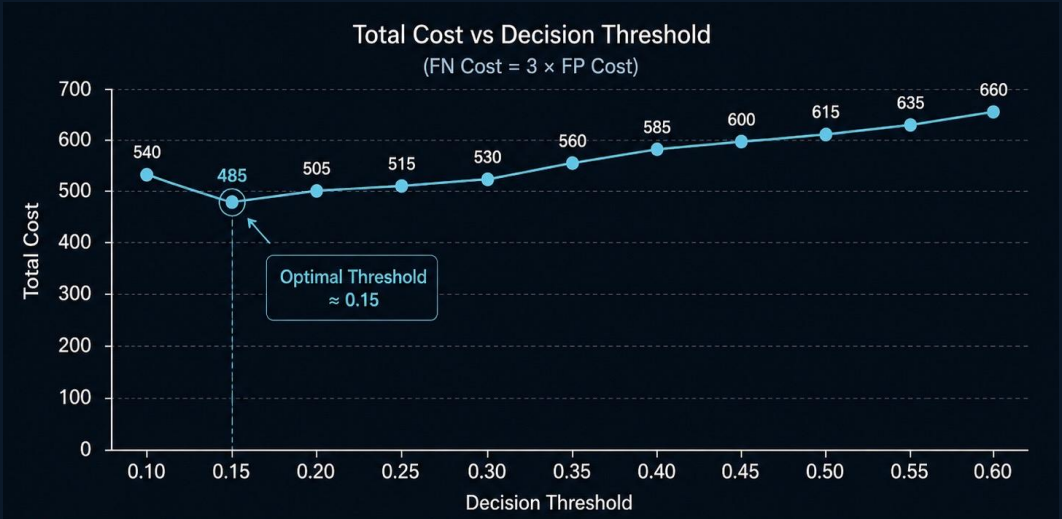
In deployment: would require knowing which season the model is being asked to predict into

Practical implication: A deployed season-aware system would require a seasonal classifier or calendar-based switching logic to select the appropriate model version. This is operationally feasible but adds system complexity relative to a single static model.

Exploratory Cost-Sensitive Decision Analysis

Illustrative threshold optimisation for methodological insight only

⚠️ This analysis uses the random-split model (AUC 0.903) and an assumed 1:3 cost ratio (FP:FN). Under temporal validation, the model performs at near-random levels. This must be treated as an illustrative of asymmetric cost principles NOT as a deployment-ready recommendation.



Interpretation: Total cost is minimized at a low decision threshold (~0.15), meaning the model should favor early positive predictions to avoid costly false negatives under the assumed cost ratio.

τ* = 0.12

Optimal Threshold

Well below the conventional 0.50 default, favouring recall

≈ 1.00

Recall at τ*

Nearly all delays captured under assumed cost structure

≈ 0.80

Precision at τ*

~20% false alarm rate, operationally manageable

−20%

Cost Reduction

vs τ = 0.50 default under assumed 1:3 cost ratio

Research Questions: Answered

Four questions: four empirically grounded answers

RQ1

SUPPORTED

XGBoost AUC 0.903 under random-split evaluation, exceeding the 0.88 benchmark. Historical delay propagation dominates feature importance (40.6%), weather secondary (33.9%). Ensemble methods outperform logistic regression by 14–16% AUC. Results hold only under random-split conditions.

RQ2

NOT SUPPORTED

Random-split AUC 0.903 collapses to 0.474 under temporal validation. Overall delay rates remain stable (79.5% → 77.6%), confirming this is a feature-relationship failure, not a base-rate change. Predictive relationships do not generalise across seasonal periods.

RQ3

MECHANISM IDENTIFIED

Primary pattern: seasonal regime shift in delay propagation (+7.07pp → −7.78pp across seasons, total swing 14.84pp). Best mitigation: propagation feature removal within season-aware framework → AUC 0.624. Causal attribution remains uncertain without delay-cause codes.

RQ4

NOT SUPPORTED

$r = 0.03$, $\Delta AUC = -0.002$, SHAP = 8.9% (lowest category). Airside operational factors dominate. Caveat: traffic data uses averaged circadian profiles; a test using actual daily traffic data remains an open question.

Study Limitations

Boundaries that should be considered when interpreting these findings

Single Airport & Single Seasonal Transition

The study covers one airport (IST) and one autumn-to-winter transition (Oct–Feb 2025–2026). Whether the observed seasonal regime shift is consistent across years, airports, or other seasonal transitions cannot be determined from this dataset alone.

Data Attrition and Unvalidated MCAR Assumption

77% of raw API records were excluded (primarily lacking actual departure times). No formal statistical test of the Missing Completely At Random assumption was conducted. If missingness is systematically related to delay status, the analytical sample may not be representative.

Surface Traffic: Averaged Profiles, Not Actual Conditions

Traffic features use 24-hour averaged circadian profiles due to data access constraints. RQ4 therefore tests whether typical circadian traffic patterns correlate with delays, not whether real-time or day-specific traffic conditions contribute predictive value.

Small Temporal Test Set

The temporal test set contains only 433 flights (Jan–Feb 2026). AUC estimates on small samples carry greater uncertainty. No confidence intervals are reported for temporal validation AUC estimates, which is a methodological limitation.

Absence of Delay-Cause Codes

The mechanisms proposed to explain the seasonal propagation reversal (e.g., 'weather-driven independence') are interpretive hypotheses. Without IATA delay-cause codes or operational records, causal attribution cannot be verified.

No Cross-Validation for Temporal Hyperparameter Selection

Standard k-fold cross-validation was used for hyperparameter tuning within the training period. This may select configurations that benefit from within-period autocorrelation, potentially introducing a mild optimism bias in the random-split results.

Thesis Contributions

Five empirical and two methodological contributions to the flight delay prediction literature

01 **Temporal Validation Evidence** — Empirical

Demonstrates that random-split AUC substantially overestimates deployment-realistic performance at a major hub airport, quantifying a gap the literature had not previously measured.

02 **Seasonal Regime Shift in Propagation** — Empirical

Documents a reversal in delay contagion dynamics across autumn and winter (−14.84pp swing), consistent with seasonal concept drift in the primary predictive mechanism.

03 **Airline-Specific Propagation Analysis** — Empirical

Quantifies heterogeneous contagion effects across carriers at IST, ranging from +21pp (TK) to +59.8pp (Aegean), providing carrier-level insight absent from prior single-airport studies.

04 **Surface Traffic Null Result** — Empirical

Provides an empirical assessment of road traffic data as a predictor, finding negligible predictive value under the available data. Establishes a baseline for future work using real-time traffic feeds.

05 **Temporal Validation Framework** — Methodological

Implements and documents a multi-month chronological validation design with strict leakage prevention, offering a replicable methodological template for future aviation ML studies.

Implications for Research and Practice

Conclusions about deployment and methodology stated within the scope of evidence

Results are based on one airport and one seasonal transition. The following implications are conditional on replication at other airports and time periods. They should be read as research-informed directions, not prescriptive recommendations.

For Researchers

- Report both random-split AND temporal-split AUC, one without the other is misleading for deployment contexts
- Multi-month chronological validation should be considered standard practice
- Investigate delay-cause codes to test propagation reversal mechanisms empirically
- Extend temporal analysis to full annual cycles and multi-airport networks
- Test real-time traffic data (not averaged profiles) before concluding traffic is uninformative

For System Designers

- Static models may be insufficient for long-term hub-airport deployment, seasonal retraining pipelines warrant consideration
- Season-aware modelling strategies may improve robustness (AUC +0.15 observed), pending further validation
- Model monitoring for concept drift is advisable before operational use; degradation can be severe (AUC < 0.5)
- Prioritise airside operational data quality over landside traffic integration based on available evidence

For Airport Operations

- Results suggest operational factors (propagation, weather) are the most informative predictors consistent with domain knowledge
- Carrier-level heterogeneity in delay contagion may inform airline-specific buffer strategies
- Any decision-support system should include explicit uncertainty quantification given temporal robustness limitations
- Seasonal transitions represent periods of elevated model unreliability, manual oversight may be advisable

Final Conclusions

Four main findings, stated within the limitations of this study

1

Strong Predictive Performance Is Achievable Under Stable Conditions

AUC 0.903 demonstrates that multi-source ML integration works when training and test distributions are aligned. Results are consistent with published benchmarks and confirm historical delay propagation and weather as the dominant predictors.

2

Temporal Robustness Represents a Critical Unresolved Challenge

The 0.903 → 0.474 AUC collapse suggests existing literature's random-split benchmarks may substantially overestimate real deployment effectiveness. Temporal robustness degradation was severe and consistent across all three evaluated architectures.

3

Seasonal Changes in Delay Propagation Are Consistent with the Observed Degradation

Delay contagion reverses from +7.07pp to −7.78pp across the study period. Features dominating training-period predictions appear to become adversarial in winter. The underlying mechanism remains a hypothesis pending verification with delay-cause data.

4

Season-Aware Modelling Offers a Promising but Incomplete Path Forward

Propagation feature removal within a season-aware framework improved winter AUC from 0.474 to 0.624. This represents meaningful improvement, though temporal robustness remains modest. Online learning, continuous retraining, and richer feature sets may be necessary for sustained operational reliability.

"Strong offline performance is necessary but not sufficient. Temporal robustness determines real-world value — and remains an open challenge."

Thank You

Questions & Discussion

Sheraz Lyna Touhouche · MSc Big Data Analytics & Management · Bahçeşehir University

Supervisor: Assoc. Prof. Tevfik Aytekin · Istanbul, June 2026